

LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 2
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Date: 08-09-2026

1. TITLE

Design and Implementation of College Information Portal using HTML and CSS

2. AIM

To develop a styled, visual college information portal webpage using HTML5 semantic elements and Cascading Style Sheets (CSS).

3. OBJECTIVE

The objectives of this experiment are:

- To apply external CSS styling to HTML elements including headers, navigation bars, cards, tables, and forms.
- To implement layout techniques like CSS Flexbox for card structures.
- To apply hover effects, borders, colors, and shadows for enhanced user visual experience.

4. THEORY

Cascading Style Sheets (CSS) describe how HTML elements are displayed on screen. Selectors, color properties, box model (margins, padding, borders), Flexbox layout (`display: flex`), and pseudo-classes (like `:hover`) enable structured, modern, and visually engaging web design.

5. ALGORITHM / PROCEDURE

1. Create an `index.html` file with semantic HTML tags (`<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<table>`, `<form>`, `<footer>`).
2. Link the external CSS file `style.css` using the `<link>` tag in the HTML head.
3. Apply global resets (`* { margin: 0; padding: 0; box-sizing: border-box; }`) in `style.css`.
4. Style the header and navigation bar with custom background colors, padding, and hover states for links.
5. Use CSS Flexbox (`display: flex`) for styling course cards with hover transform effects.
6. Style the facilities table with dark headers, alternating row borders, and centered text.

7. Style form input fields, textarea, and submit button with custom colors, rounded corners, and hover effects.
8. Verify rendering in a browser.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>

  <!-- Link CSS file -->
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- Header -->
  <header>
    <h1>ANITS College </h1>
    <p>College Information Portal</p>
  </header>

  <!-- Navigation -->
  <nav>
    <a href="#home">Home</a>
    <a href="#about">About</a>
    <a href="#courses">Courses</a>
    <a href="#contact">Contact</a>
  </nav>

  <!-- Main Content -->
  <main>

    <!-- Home -->
    <section id="home">
      <h2>Welcome</h2>
      <p>
        Welcome to ANITS College.
        We provide quality education and technical skills
        to students.
      </p>
    </section>

    <!-- About -->
    <section id="about">
      <h2>About Us</h2>
```

```

    <p>
        Our college provides excellent education,
        modern laboratories and experienced faculty.
    </p>
</section>

<!-- Courses -->
<section id="courses">
    <h2>Courses Offered</h2>

    <div class="courses">

        <article>
            <h3>Computer Science</h3>
            <p>B.Tech CSE</p>
        </article>

        <article>
            <h3>Information Technology</h3>
            <p>B.Tech IT</p>
        </article>

        <article>
            <h3>Electronics</h3>
            <p>B.Tech ECE</p>
        </article>

    </div>
</section>

<!-- Facilities -->
<section>
    <h2>College Facilities</h2>

    <table>
        <tr>
            <th>Facility</th>
            <th>Status</th>
        </tr>

        <tr>
            <td>Library</td>
            <td>Available</td>
        </tr>

        <tr>
            <td>Computer Lab</td>
            <td>Available</td>
        </tr>

        <tr>
            <td>Sports Ground</td>
            <td>Available</td>
        </tr>
    </table>

```

```

        </table>
    </section>

    <!-- Contact -->
    <section id="contact">
        <h2>Contact Us</h2>

        <form>
            <label>Name:</label>
            <input type="text" placeholder="Enter your name">

            <label>Email:</label>
            <input type="email" placeholder="Enter your email">

            <label>Message:</label>
            <textarea placeholder="Enter your message"></textarea>

            <input type="submit" value="Submit">
        </form>
    </section>

</main>

<!-- Footer -->
<footer>
    <p>&copy; 2026 ABC College of Engineering</p>
</footer>

</body>
</html>

```

style.css

```

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

/* Body */
body {
    font-family: Arial, sans-serif;
    background-color: #d79c9c;
    color: #4f7f9f;
}

/* Header */
header {
    background-color: #2c3e50;
    color: rgba(56, 94, 35, 0.662);
    text-align: center;
    padding: 30px;
}

```

```

header h1 {
    font-size: 35px;
}

header p {
    margin-top: 10px;
    font-size: 18px;
}

/* Navigation */
nav {
    background-color: #8d3590;
    text-align: center;
    padding: 15px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 20px;
    font-size: 18px;
}

/* Hover Effect */
nav a:hover {
    color: yellow;
}

/* Main */
main {
    width: 90%;
    margin: 25px auto;
}

/* Sections */
section {
    background-color: white;
    padding: 25px;
    margin-bottom: 20px;
    border-radius: 10px;
    box-shadow: 0 2px 8px #ccc;
}

section h2 {
    color: #2c3e50;
    margin-bottom: 15px;
}

/* Courses */
.courses {
    display: flex;
    gap: 20px;
    flex-wrap: wrap;

```

```

}

article {
    background-color: #3498db;
    color: white;
    padding: 20px;
    text-align: center;
    flex: 1;
    border-radius: 8px;
}

/* Article Hover */
article:hover {
    transform: scale(1.05);
    transition: 0.3s;
}

/* Table */
table {
    width: 100%;
    border-collapse: collapse;
}

th, td {
    border: 1px solid #ccc;
    padding: 12px;
    text-align: center;
}

th {
    background-color: #34495e;
    color: white;
}

/* Form */
input, textarea {
    width: 100%;
    padding: 10px;
    margin: 8px 0 15px;
    border: 1px solid #ccc;
    border-radius: 5px;
}

textarea {
    height: 100px;
}

/* Submit Button */
input[type="submit"] {
    background-color: #27ae60;
    color: white;
    border: none;
    cursor: pointer;
    font-size: 16px;
}

```

```

}

input[type="submit"]:hover {
    background-color: #219150;
}

/* Footer */
footer {
    background-color: #2c3e50;
    color: white;
    text-align: center;
    padding: 20px;
}

```

7. CODE EXPLANATION

Code / Syntax	Explanation
<code>display: flex; gap: 20px;</code>	Uses Flexbox layout to align course cards side-by-side with equal spacing.
<code>article:hover { transform: scale(1.05); }</code>	Scales course card slightly on hover for smooth interactive animation.
<code>box-shadow: 0 2px 8px #ccc;</code>	Adds drop shadow around sections for visual depth.
<code>input[type="submit"]</code>	Styles submit button with custom background, border-radius, and hover transition.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	CSS Link	Load index.html in browser	Styles applied from style.css	Pass
TC-02	Flexbox Layout	View Courses section	Cards arranged horizontally in flex row	Pass
TC-03	Navigation Hover	Hover cursor over nav links	Link text turns yellow	Pass

9. OUTPUT

ANITS College

College Information Portal

[Home](#) [About](#) [Courses](#) [Contact](#)

Welcome

Welcome to ANITS College. We provide quality education and technical skills to students.

About Us

Our college provides excellent education, modern laboratories and experienced faculty.

Courses Offered

Computer Science
B.Tech CSE

Information Technology
B.Tech IT

Electronics
B.Tech ECE

College Facilities

Facility	Status
Library	Available
Computer Lab	Available
Sports Ground	Available

Contact Us

Name:

Email:

Message:

Submit

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10. RESULT

Thus, the college information portal webpage demonstrating CSS styling, layout handling, and visual enhancements was successfully implemented and verified.